



Course Title: Microcomputer Application I
Course Code: CSE-2402

Topic: Microsoft Excel topics
(Basic idea, Cell formatting, Equations, and Formulas)

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Lecture Agenda

Three core topics you will master by end of this lecture



01

Basic Idea of MS Excel

What it is, the interface, rows, columns & cells

02

Cell Formatting

Number formats, fonts, borders, colours, alignment

03

Equations & Formulas

Writing formulas, built-in functions, references

04

Worked Examples

Real spreadsheet scenarios with step-by-step formulas

05

Practice Questions

6 questions across all three topics

06

Lab Task

Build a Student Marks Calculator — 45 min task

What is Microsoft Excel?



Topic 01 — Basic Idea

Definition:

Microsoft Excel is a spreadsheet application used to store, organise, analyse and visualise data using rows, columns and formulas.



Data Analysis

Sort, filter and summarise large datasets quickly



Calculations

Perform complex maths, statistics and financial modelling



Charts & Graphs

Visualise trends with bar, line, pie and scatter charts



Record Keeping

Manage budgets, inventories, marks sheets and schedules

★ 1,048,576 rows per sheet

★ 16,384 columns per sheet

★ Columns labelled A–Z, AA–XFD

★ Rows labelled 1–1,048,576

File Formats:



.xlsx

Modern Excel
(default)

.xls

Legacy Excel
97-2003

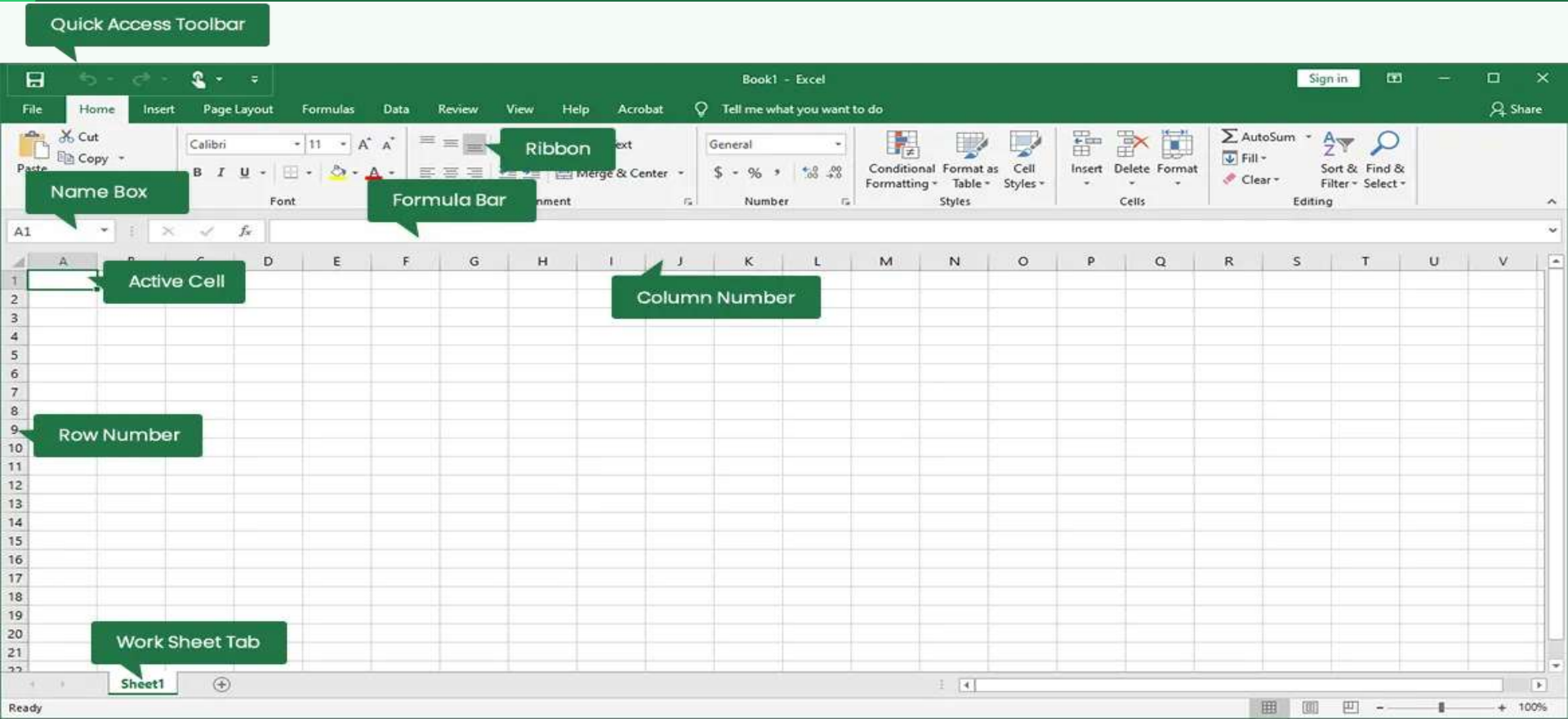
.csv

Comma-Separated
Values

.xlsm

Excel with Macros

The Excel Interface — Labelled Diagram



Rows, Columns & Cell References

Understanding the Excel grid structure



COLUMN (Vertical — labelled A, B, C...)

	A	B	C	D	E
--	---	---	---	---	---

ROW
(Horizontal
1, 2,
3...)

1	Name	Score	Grade		
2	Ali	85			
3	Sara	72			
4	Omar	91			
5	Nadia	60			

Cell Reference

Each cell has a unique address: Column letter + Row number.
Example: C3 means Column C, Row 3

Range

A range is a block of cells. Written as Start:End.
Example: B2:B5 means cells B2, B3, B4 and B5

Active Cell

The selected cell — shown with a green border and its address in the Name Box (top-left).

Relative Cell Reference (B3)

Changes automatically when you copy the formula to another cell.

E.g. =A3+B3

Absolute Cell Reference (\$B\$3)

Remains fixed, no matter where the formula is copied.

E.g. =A3*\$B\$3

Cell Formatting

Topic 02 — Making your spreadsheet clear and professional



Number Format

Controls how numbers appear:
General, Number, Currency, Percentage, Date,
Text, Scientific



Font & Text

Font face, size, bold, italic, underline, colour and
special effects (strikethrough, superscript)



Fill Colour

Background colour of a cell or range. Use
consistently to highlight headers, totals or
categories



Borders

Add lines around or inside cells. Choose thickness,
style (solid/dashed) and colour



Alignment

Horizontal (left/centre/right) and vertical
(top/middle/bottom). Also: Wrap Text & Merge
Cells



Format Painter

Copy formatting from one cell to another in one
click. Double-click to apply to multiple cells

Cell Formatting — Before & After Examples

See the difference formatting makes



BEFORE (No Formatting)			
	Product	Price	Stock
1	Laptop	1200	45
2	Mouse	25	200
3	Keyboard	75	130
4	Monitor	350	60
5	Headset	90	88
		1740	523

Format
Applied

AFTER (Formatted)			
	Product	Price	In Stock
1	Laptop	\$1,200.00	45
2	Mouse	\$25.00	200
3	Keyboard	\$75.00	130
4	Monitor	\$350.00	60
5	Headset	\$90.00	88
	TOTAL	\$1,740.00	523

Currency format

Numbers displayed as \$1,200.00

Bold + Green header

Header row styled with fill colour

Alternating rows

Easier to read with light green fill

Bold totals row

Total row stands out clearly

Number Formats — Same Number, Different Display



Topic 02 continued — Format Cells dialog

Key insight:

The underlying value stored in a cell never changes when you apply a Number Format — only the way it is displayed changes.

Format Type	Raw Value	Displayed As	Example Use	How to Apply
General	1234.5	1234.5	Default — no special format	Ctrl+Shift+~
Number	1234.5	1,234.50	Quantities with decimals	Ctrl+Shift+1
Currency	1234.5	\$1,234.50	Prices, invoices, budgets	Ctrl+Shift+\$
Percentage	0.875	87.5%	Scores, growth rates	Ctrl+Shift+%
Date	45383	01/01/2024	Dates, schedules	Ctrl+Shift+#
Time	0.5	12:00 PM	Schedules, durations	Ctrl+Shift+@
Scientific	1234500	1.23E+06	Very large/small numbers	Ctrl+Shift+^
Text	001234	001234	IDs, codes, phone numbers	Ctrl+Shift+~

💡 *Shortcut: Select cells → Right-click → Format Cells (Ctrl+1) → Number tab → choose format and set decimal places.*

Introduction to Formulas & Equations



Topic 03 — The power of Excel

Golden Rule:

Every formula in Excel must begin with an equals sign(=), without it, Excel treats your entry as plain text.

Anatomy of a Formula:

= SUM (B2 : B10) + C2 * 0.1

Equals sign (required)	Function name	Cell range (argument)	Cell reference	Operator * (multiply)	Constant value
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Arithmetic Operators:

Op.	Meaning	Example Formula	Op.	Meaning	Example Formula	BODMAS Rule
+	Addition	=A1+B1	%	Percentage	=A1*10%	
-	Subtraction	=A1-B1	&	Concatenate	=A1&" "&B1	
*	Multiplication	=A1*B1	=	Equals	=A1=B1	
/	Division	=A1/B1	>	Greater Than	=A1>100	
^	Exponent	=A1^2				
						1. Brackets first 2. Orders (^) 3. Division 4. Multiplication 5. Addition 6. Subtraction

Essential Built-In Functions



Topic 03 — Excel has 400+ functions — here are the most important ones

SUM

`=SUM(B2:B10)`

= 308

Adds all numbers in a range

e.g. Total of sales column

AVERAGE

`=AVERAGE(B2:B10)`

= 34.2

Calculates the arithmetic mean

e.g. Average exam score

MAX

`=MAX(B2:B10)`

= 91

Returns the largest value in a range

e.g. Highest mark in class

MIN

`=MIN(B2:B10)`

= 22

Returns the smallest value in a range

e.g. Lowest temperature recorded

COUNT

`=COUNT(B2:B10)`

= 9

Counts cells that contain numbers

e.g. How many students have marks

COUNTA

`=COUNTA(A2:A10)`

= 9

Counts all non-empty cells

e.g. How many names are filled in

IF

`=IF(B2>=50,"Pass","Fail")`

= Pass

Returns one of two values based on condition

e.g. Grade pass/fail

ROUND

`=ROUND(B2,2)`

= 85.47

Rounds a number to specified decimal places

e.g. Round currency values

💡 Type = then the function name and Excel shows an autocomplete suggestion with the full syntax guide automatically.

Worked Example — Student Marks Calculator

See all three topics applied together



Student Marks Calculator

	Name	Quiz	Mid	Final	Total	Grade
1	Ali	18	22	45	=C3+D3+E3	Pass
2	Sara	14	19	38	=C4+D4+E4	Pass
3	Omar	20	25	48	=C5+D5+E5	Pass
4	Nadia	12	17	32	=C6+D6+E6	Fail
5	Khalid	17	21	41	=C7+D7+E7	Pass
	MAX	=MAX(C3:C7)	=MAX(D3:D7)	=MAX(E3:E7)	=MAX(F3:F7)	
	AVG	=AVG(C3:C7)	=AVG(D3:D7)	=AVG(E3:E7)	=AVG(F3:F7)	

Grade formula in G3: =IF(F3>=50,"Pass","Fail") — then drag down to G7

1

Design the grid

Set up columns: Name, Quiz (20), Midterm (25), Final (50), Total, Grade. Format header row green & bold.

2

Enter data

Type student names and raw marks. Use Number format on score columns, 0 decimal places.

3

Write Total formula

In F3: =C3+D3+E3 then drag down to F7. This adds all three marks for each student.

4

Write Grade formula

In G3: =IF(F3>=50,"Pass","Fail") — tests if total is 50 or above and returns Pass or Fail.

5

MAX & AVERAGE rows

In F8: =MAX(F3:F7) for highest total. In F9: =AVERAGE(F3:F7) for class average.



Always use cell references (=C3+D3+E3) instead of typing numbers (=18+22+45) — if marks change, totals update automatically!

Common Formula Errors & How to Fix Them



Don't panic — every error has a clear cause and solution

#DIV/0!

Cause: Dividing by zero or empty cell

Fix: Check denominator is not 0 or blank. Use `=IF(B2=0,0,A2/B2)` to handle it safely.

#VALUE!

Cause: Wrong data type in formula

Fix: Check that all cells contain numbers, not text. Use `VALUE()` to convert text to numbers.

#REF!

Cause: Formula refers to a deleted cell/range

Fix: Undo the deletion (Ctrl+Z) or rewrite the formula with the correct cell reference.

#NAME?

Cause: Excel does not recognise the function

Fix: Check spelling — SUUM instead of SUM. Function names are not case-sensitive but must be spelled correctly.

#N/A

Cause: Value not found (e.g. VLOOKUP)

Fix: The lookup value does not exist in the table. Check spelling or use `IFERROR()` to handle it.

#NUM!

Cause: Invalid numeric value in formula

Fix: Check for impossible maths, e.g. SQRT of a negative number or a number too large for Excel.

#####

Cause: Column too narrow to display the value

Fix: Simply double-click the column border to auto-fit the column width. Not a real error.

Circular

Cause: Formula refers back to its own cell

Fix: A cell cannot calculate itself. Excel will warn you. Trace dependents to find the loop.

💡 Use Formulas tab → Error Checking to scan your sheet automatically, or hover over the green triangle on an error cell for a quick fix suggestion.

Practice Questions

Test your understanding across all three topics



Q1

MCQ

Which of the following is a valid cell address in Excel?

- A) 1A B) AA C) B12 D) 12B

Q2

Short Answer

What is the difference between a Relative cell reference (B3) and an Absolute cell reference (\$B\$3)? Give one example of when you would use each.

Q3

Formula

Write a formula for cell F2 that calculates the average of cells B2 to E2, then rounds the result to 1 decimal place.

Q4

MCQ

A student types 1234 in cell A1 and applies Currency format. What will the cell display?

- A) 1234 B) \$1,234.00 C) 1234.00 D) \$1234

Q5

Short Answer

A cell shows the error #DIV/0! — explain what caused this error and write an IF formula that would prevent it when dividing A2 by B2.

Q6

Formula

A sales table has Product names in A2:A10 and Revenue in B2:B10. Write the formulas to find: (i) Total revenue (ii) Highest single sale (iii) Count of sales entries



Lab Task — Hands-On Practice



Duration: 40–45 minutes | Individual | Submit as Firstname_Lastname_Excel_Lab.xlsx

Scenario:

You are a teacher's assistant. Build a Student Performance Tracker spreadsheet for a class of 8 students across 4 subjects, using all the skills learned today.

Data Setup

- ◆ Row 1: Headers — Name, Maths, Science, English, IT
- ◆ Add 8 student names in A2:A9
- ◆ Enter marks (0–100) for all 4 subjects
- ◆ Add a row 11 labelled 'Class Average'
- ◆ Add a row 12 labelled 'Highest Mark'
- ◆ Add a row 13 labelled 'Lowest Mark'

Formulas Required

- ◆ Column F (Total): =SUM(B2:E2) — copy to all 8 students
- ◆ Column G (Average): =AVERAGE(B2:E2) — copy down
- ◆ Column H (Grade): IF Average>=90=A, >=75=B, >=60=C, else=D
- ◆ Row 11: AVERAGE of each subject column (B–F)
- ◆ Row 12: MAX of each subject column (B–F)
- ◆ Row 13: MIN of each subject column (B–F)

Formatting Required

- ◆ Header row: Bold, Green fill, White text
- ◆ Score columns (B–E): Number format, 0 decimal places
- ◆ Average column (G): Number format, 1 decimal place
- ◆ Rows 11–13: Bold, Light green fill
- ◆ All cells: Borders on all sides
- ◆ Column widths: Auto-fit all columns (double-click borders)



Marking: Data Setup 20 pts + Formulas 50 pts + Formatting 30 pts = 100 pts

Thank You

Any Questions?